

Erdős Problem #359

EP-359

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $a_1 < a_2 < \dots$ be an infinite sequence of integers such that $a_1 = n$ and a_{i+1} is the least integer which is not a sum of consecutive earlier a_j s. What can be said about the density of this sequence?

In particular, in the case $n = 1$, can one prove that $a_k/k \rightarrow \infty$ and $a_k/k^{1+c} \rightarrow 0$ for any $c > 0$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>